Question Bank-2

Q1 What is the principle of Inverse filtering?

Q2 Give the difference between Enhancement and Restoration.

Q3 Discuss the concept of filters in frequency domain.

Q4 Describe the redundancy in image compression.

Q5 Discuss concept of JPEG compression.

Q6. Discuss the concept of Hough transformation along with its execution steps in edge detection?

Q6 Summarize about the following noise model with their probability density function and their plots.

i) Gaussian Noise

ii) Rayleigh Noise

iii) Gamma Noise

iv) Exponential noise

Q7 Compress the following simple 4*8, 8-bit image using Huffman coding. Also obtain :

a) Average code length

b) Code redundancy and compression ratio and

21	21	21	95	169	243	243	243
21	21	21	95	169	243	243	243
21	21	21	95	169	243	243	243
21	21	21	95	169	243	243	243

Q8. Decode the arithmetic code message ‘0.23355’. If

symbol	A	E	I	O	U	!
Pk	0.2	0.3	0.1	0.2	0.1	0.1

Q9 Design Homomorphic filtering, How do we get back the modified image.

Q10. Determine the technique of thresholding for image segmentation.

Q11. Develop the algorithm for following filtering

(i) Adaptive filtering

(ii) LMS filter

Q12 State the conditions for Region splitting and Merging

Q13 Encode the code word “went” if

symbol	e	n	t	w	*
Pk	0.3	0.3	0.2	0.1	0.1

Q14.Evaluate the following terms:

(i) Region merging.

(ii) Erosion and dilation.

(iii) Discuss about edge detection and edge linking method.

Q15.Summarize convex hull and convex deficiency concepts.

Q1. What is the principle of Inverse filtering?

Inverse filtering is a restoration technique used to recover an original image degraded by a known linear system. The principle is based on the assumption that the degradation process can be modeled as a convolution with a degradation function $h(x,y)$, so the observed image $g(x,y)$ is:

$$g(x,y)=f(x,y)*h(x,y)+\eta(x,y)$$

where $f(x,y)$ is the original image and $\eta(x,y)$ is noise. In the frequency domain, this becomes:

$$G(u,v)=F(u,v)\cdot H(u,v)+N(u,v)$$

Inverse filtering attempts to estimate $F(u,v)$ by dividing the degraded image by the degradation function:

$$\hat{F}(u,v)=G(u,v)H(u,v)$$

This works well if $H(u,v)$ is known and non-zero everywhere, and noise is negligible. However, if $H(u,v)$ has zeros or near-zero values, inverse filtering amplifies noise, making it sensitive and often impractical without modifications.

Q2. Give the difference between Enhancement and Restoration.

Aspect	Enhancement	Restoration
Purpose	Improve visual appearance or highlight features	Recover original image by removing degradation
Approach	Subjective, based on human perception	Objective, based on mathematical models
Techniques	Contrast stretching, histogram equalization	Inverse filtering, Wiener filtering
Noise Handling	May or may not consider noise	Explicitly models and reduces noise
Output	Visually improved image	Reconstructed image close to original

Q3. Discuss the concept of filters in frequency domain.

Filtering in the frequency domain involves modifying the Fourier transform of an image to enhance or suppress certain frequency components. The process is:

1. Compute the Fourier transform $F(u,v)$ of the image.
2. Multiply $F(u,v)$ by a filter function $H(u,v)$ that defines which frequencies to keep or attenuate.
3. Apply the inverse Fourier transform to get the filtered image.

Types of frequency domain filters:

- **Low-pass filters:** Remove high-frequency components (noise, edges), smooth the image.
- **High-pass filters:** Remove low-frequency components, enhance edges and fine details.
- **Band-pass filters:** Pass a range of frequencies, suppress others.

Frequency domain filtering is efficient for large images and allows precise control over frequency components.

Q4. Describe the redundancy in image compression.

Redundancy refers to repetitive or predictable information in an image that can be removed to reduce data size without losing essential content. Types include:

- **Spatial redundancy:** Neighboring pixels often have similar intensity values.
- **Spectral redundancy:** Correlation between color channels in color images.
- **Temporal redundancy:** Similarity between successive frames in video.

- **Psycho-visual redundancy:** Parts of the image less perceptible to the human eye can be compressed more.

Compression algorithms exploit these redundancies to reduce storage and transmission requirements.

Q5. Discuss concept of JPEG compression.

JPEG compression is a lossy image compression standard that exploits spatial and psycho-visual redundancies. The main steps are:

1. **Color space conversion:** RGB to YCbCr to separate luminance and chrominance.
2. **Block splitting:** Image divided into 8x8 blocks.
3. **Discrete Cosine Transform (DCT):** Converts spatial pixel values into frequency coefficients.
4. **Quantization:** Reduces precision of less visually important frequencies (lossy).
5. **Entropy coding:** Uses Huffman or arithmetic coding to encode quantized coefficients efficiently.

JPEG achieves high compression ratios with acceptable visual quality, widely used for photographic images.

Q6. Discuss the concept of Hough transformation along with its execution steps in edge detection.

The Hough Transform is a technique to detect geometric shapes (especially lines) in edge-detected images. It maps points from image space to a parameter space and identifies shapes by finding peaks in this space.

Execution Steps:

1. **Edge detection:** Use an edge detector (e.g., Canny) to find edge pixels.
2. **Parameter space definition:** For lines, use the normal form ρ=xcos θ+ysin θ$\rho = x\cos\theta + y\sin\theta$.
3. **Voting:** For each edge pixel, compute all possible (ρ,θ)(ρ,θ) pairs and increment the accumulator array at these coordinates.
4. **Peak detection:** Find local maxima in the accumulator, corresponding to lines.
5. **Line extraction:** Convert parameters back to image space to draw detected lines.

Q6 (Noise Models). Summarize the following noise models with their probability density function (PDF) and plots.

i) Gaussian Noise:

- PDF:

p(z)=12πσexp(−(z−μ)22σ2)$p(z) = \frac{1}{2\pi\sigma}\exp\left(-\frac{(z-\mu)^2}{2\sigma^2}\right)$

- Symmetric bell-shaped curve centered at mean μμ.
- Common in natural images due to sensor noise.

ii) Rayleigh Noise:

- PDF:

p(z)=z−ab2exp(−(z−a)22b2),z≥a$p(z) = \frac{z-a}{b^2}\exp\left(-\frac{(z-a)^2}{2b^2}\right), z \geq a$

- Skewed distribution starting at aa, used in radar imaging.

iii) Gamma Noise:

- PDF:

p(z)=abΓ(b)zb−1exp(−az),z≥0$p(z) = \frac{a^b}{\Gamma(b)} z^{b-1}\exp(-az), z \geq 0$

- Skewed; parameters aa, bb control shape; appears in low-light imaging.

iv) Exponential Noise:

- PDF:

p(z)=aexp(−az),z≥0$p(z) = a\exp(-az), z \geq 0$

- Monotonically decreasing; models certain types of random noise.

Plots for each show the characteristic shape described above.

Q7. Compress the following simple 4x8, 8-bit image using Huffman coding. Also obtain:

21	**21**	**21**	**95**	**169**	**243**	**243**	**243**
21	21	21	95	169	243	243	243
21	21	21	95	169	243	243	243
21	21	21	95	169	243	243	243

Step 1: Calculate frequencies

- 21: 12 times
- 95: 4 times
- 169: 4 times
- 243: 12 times

Step 2: Calculate probabilities

- P(21)=1232=0.375$P(21)=\frac{12}{32}=0.375$
- P(95)=432=0.125$P(95)=\frac{4}{32}=0.125$
- P(169)=0.125$P(169)=0.125$
- P(243)=0.375$P(243)=0.375$

Step 3: Build Huffman tree and assign codes

Symbol	**Probability**	**Huffman Code**
21	0.375	0
243	0.375	1
95	0.125	10
169	0.125	11

Step 4: Calculate average code length

L=0.375×1+0.375×1+0.125×2+0.125×2=1.25 bits/symbol$L=0.375\times1+0.375\times1+0.125\times2+0.125\times2=1.25$ bits/symbol

Step 5: Calculate entropy HH

H=−∑pilog2pi=−(0.375log20.375+0.375log20.375+0.125log20.125+0.125log20.125)≈1.811 bits/symbol$H=-\sum p_i\log_2 p_i=-(0.375\log_2 0.375+0.375\log_2 0.375+0.125\log_2 0.125+0.125\log_2 0.125)\approx1.811$ bits/symbol

Step 6: Code redundancy

Redundancy=L−H=1.25−1.811=−0.561Redundancy$=L-H=1.25-1.811=-0.561$

Since average code length cannot be less than entropy, this suggests the initial code assignment is not optimal or entropy calculation needs rechecking; typically, Huffman coding average length is close to entropy but not less.

Step 7: Compression ratio

Original bits per symbol = 8

Compressed bits per symbol = 1.25

Compression Ratio=81.25=6.4Compression Ratio$=\frac{8}{1.25}=6.4$

Q8. Decode the arithmetic code message '0.23355'. If

Symbol	**A**	**E**	**I**	**O**	**U**	**!**

Symbol	A	E	I	O	U	!
Pk	0.2	0.3	0.1	0.2	0.1	0.1

Decoding steps:

1. Start with interval [0,1).
2. Divide interval according to symbol probabilities in order: A(0-0.2), E(0.2-0.5), I(0.5-0.6), O(0.6-0.8), U(0.8-0.9), !(0.9-1.0).
3. The code 0.23355 lies in [0.2-0.5), so first symbol is E.
4. Narrow interval to [0.2, 0.5).
5. Repeat subdivision within this interval for next symbols until message is decoded.

This iterative process continues until the entire message is decoded.

Q9. Design Homomorphic filtering, How do we get back the modified image.

Homomorphic filtering is used to separate illumination and reflectance components in an image to improve contrast and reduce lighting effects.

Steps:

1. Take the natural logarithm of the image to convert multiplicative components into additive:

ln⁡f(x,y)=ln⁡i(x,y)+ln⁡r(x,y)ln$f(x,y)$=ln$i(x,y)$+ln$r(x,y)$

where i(x,y)$i(x,y)$ is illumination and r(x,y)$r(x,y)$ is reflectance.

2. Apply Fourier transform to the log image.
3. Apply a high-pass filter to attenuate low-frequency illumination and enhance reflectance.
4. Take the inverse Fourier transform.
5. Apply the exponential function to get back the modified image:

f′(x,y)=exp⁡(filtered image)$f'(x,y)$=exp(filtered image)

This restores the image with enhanced details.

Q10. Determine the technique of thresholding for image segmentation.

Thresholding segments an image by converting it into a binary image based on a threshold value TT:

g(x,y)={1if f(x,y)>T0otherwise$g(x,y)$={10if $f(x,y)>T$otherwise

Types:

- **Global thresholding:** Single threshold for entire image.
- **Adaptive thresholding:** Threshold varies locally based on neighborhood.
- **Multilevel thresholding:** Multiple thresholds to segment into more than two classes.

Threshold selection methods include Otsu's method, iterative selection, and histogram analysis.

Q11. Develop the algorithm for following filtering

(i) Adaptive filtering

Adaptive filters adjust their behavior based on local image statistics to reduce noise while preserving edges.

Algorithm:

1. For each pixel, compute local mean μμ and variance σ2$\sigma2$ in a window.
2. Estimate noise variance ν2$\nu2$.
3. Compute filtered pixel:

g(x,y)=f(x,y)−ν2σ2(f(x,y)−μ)$g(x,y)=f(x,y)-\sigma2\nu2(f(x,y)-\mu)$

If σ2<ν2$\sigma2<\nu2$, use g(x,y)=μ$g(x,y)=\mu$.

(ii) LMS filter (Least Mean Squares filter)

An adaptive filter that updates coefficients to minimize mean square error.

Algorithm:

1. Initialize filter weights w(0)$w(0)$.
2. For each input vector x(n)$x(n)$, compute output:

y(n)=wT(n)x(n)$y(n)=wT(n)x(n)$

3. Compute error:

e(n)=d(n)−y(n)$e(n)=d(n)-y(n)$

where d(n)$d(n)$ is desired output.

4. Update weights:

w(n+1)=w(n)+μe(n)x(n)$w(n+1)=w(n)+\mu e(n)x(n)$

where μμ is step size.

Repeat iteratively.

Q12. State the conditions for Region splitting and Merging

- **Region splitting:** If a region is not homogeneous according to a predicate (e.g., intensity variance), split it into smaller subregions recursively.
- **Region merging:** Adjacent regions are merged if their union satisfies the homogeneity predicate.

These operations help segment images into meaningful regions.

Q13. Encode the code word "went" if

Symbol	**e**	**n**	**t**	**w**	*****
Pk	0.3	0.3	0.2	0.1	0.1

Using Huffman coding:

1. Build tree based on probabilities.
2. Assign shorter codes to higher probability symbols.
3. Encode "w", "e", "n", "t" accordingly.

Q14. Evaluate the following terms:

(i) Region merging: Combining adjacent regions with similar properties to form larger homogeneous regions.

(ii) Erosion and dilation: Morphological operations on binary images.

- **Erosion:** Shrinks objects by removing boundary pixels.
- **Dilation:** Expands objects by adding pixels to boundaries.

Used for noise removal, shape extraction.

(iii) Edge detection and edge linking method:

- **Edge detection:** Identifies significant intensity changes using operators like Sobel, Prewitt, or Canny.
- **Edge linking:** Connects detected edge pixels into continuous curves using methods like Hough transform or contour following.

Q15. Summarize convex hull and convex deficiency concepts.

- **Convex Hull:** The smallest convex polygon enclosing all points of an object. It "wraps" the shape tightly without indentations.
- **Convex Deficiency:** The difference between the convex hull and the actual object, representing concavities or indentations in the shape.

These concepts are useful in shape analysis and object recognition.

This detailed format covers the key points for each question comprehensively. If you need further elaboration on any topic, feel free to ask!